

16.07 Dynamics — Quiz 2: Foucault's Pendulum

Name: _____

1. What is the precession period of a Foucault pendulum placed at the North Pole? State your answer in sidereal days.
2. If we double the length of a Foucault pendulum, what happens to its swing period? What happens to its precession period?
3. The linearized equations of motion of the Foucault pendulum have the form $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$, where $\mathbf{C}$ is skew-symmetric ($\mathbf{C}^T = -\mathbf{C}$). What is the work done by the Coriolis force $-\mathbf{C}\dot{\mathbf{q}}$? Is the energy of the Foucault pendulum conserved?